IDA117V · INTRODUCTION TO DATA SCIENCE · GROUP PROJECT

Predicting **municipal audit outcomes** a year ahead

Helping National Treasury reach struggling municipalities before their next Auditor-General audit

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Lecturer: Dr Dan Masethe · 20 October 2026

01 · THE PROBLEM

Audit results arrive too late to prevent the damage

The problem

The Auditor-General audits all 257 municipalities every year. Poor financial management shows up as unpaid bills, neglected infrastructure and failing water, electricity and refuse services, but the audit result only arrives after the year has closed.

Who decides

National Treasury's local government support function and the provincial treasuries choose where to send finance secondments, oversight and support each year, with a limited budget and staff.

Public value

Earlier, consistent, evidence-based targeting of support, reaching struggling municipalities before service delivery collapses. The model supports human judgement; it does not replace the audit.

02 · THE TASK

One municipality, one year: predict next year's audit

INPUTS · YEAR T

9 audited financial indicators: revenue, expenditure, surplus, cash coverage, collection rate, debtor days, repairs, irregular spending, grants

+ last year's audit outcome



TARGET · YEAR T+1

At risk: qualified, adverse, disclaimer or outstanding

Not at risk: unqualified, with or without findings

Primary metric: macro-F1

Both classes matter equally; accuracy alone hides missed cases.

A missed municipality

Support arrives a year late, after services have already failed.

A false alarm

Scarce support and staff go where they were not needed.

03 · THE DATA

National Treasury's own data, free to reuse

257
municipalities

1,434
municipality-years

2018–23
feature years

45.5%
at risk next year

Source and licence

Municipal Money (municipaldata.treasury.gov.za): income and expenditure, financial position, debtors, repairs, irregular expenditure, grants and Auditor-General opinions. Free commercial and non-commercial reuse.

Privacy and quality

No personal data (POPIA): officials' names and contacts excluded. Self-reported figures include implausible values; early missing years are stored as zero; the grants extract is incomplete.

04 · LEAKAGE-SAFE SPLIT

Split by time, so the model only learns from the past

Train · 2018–2021
950 rows · fit and tune

Validate 2022
244 · pick champion

Test 2023
240 · used once

Forecast 2024
245 · 2025 audit

Rules we kept

- Imputation, scaling and encoding learned on training years only
- Year, identifiers and raw opinion columns excluded
- Tuning scored by walk-forward CV inside 2018–2021

A correction we made

Our first run used a random split and the year as a feature, letting the future inform the past. We re-ran every model on the time split; the champion changed from Random Forest to Logistic Regression.

Two baselines and four models on the same 2022 municipalities

Model	Family	Macro-F1	ROC-AUC	CV macro-F1
Dummy (majority class)	Baseline	0.366	0.500	0.33 ± 0.03
Repeat last year's outcome	Baseline	0.887	0.887	0.87 ± 0.02
Logistic Regression	Traditional	**0.887**	0.928	**0.87 ± 0.02**
Random Forest (200 trees)	Traditional	0.878	0.913	0.87 ± 0.02
MLP (64–32, dropout 0.2)	Neural	0.879	0.929	0.75 ± 0.18
Residual tabular network	Neural	0.879	0.927	0.74 ± 0.20

Champion: Logistic Regression, chosen on validation only: best macro-F1 and PR-AUC (0.922) among the four models, and the most stable across walk-forward folds.

06 · WHAT THE MODELS LEARNED

Audit outcomes repeat, and the models learned exactly that

100%

of Logistic Regression's validation predictions match “repeat last year's outcome”

97–98%

for Random Forest and both neural networks

0.887

macro-F1 of the rule alone, equal to the best model

07 · OVERFITTING AND OPTIMISATION

Tuning made the champion simpler, not more accurate

Overfitting check

1.00 → 0.88

Random Forest memorises the training years (train vs validation macro-F1).

Logistic Regression and both networks show no gap. Early stopping halted the networks at epochs 13 and 8, with dropout and weight decay as further controls.

Optuna, 60 trials

0.870 → 0.874

CV macro-F1 before and after tuning: smaller than the fold-to-fold spread (± 0.025), so not a meaningful gain.

Chosen: L1 penalty, C = 0.074. It switched off 5 of the 9 ratios, keeping cash coverage, repairs and maintenance, operating surplus and revenue.

08 · ONE-TIME TEST

Scored once on unseen 2023 figures: 90.8% correct

0.906

Macro-F1

95% CI 0.866–0.941

90/103

Failing caught

recall 0.874 · 13 missed

9/137

False alarms

precision 0.909

0.896

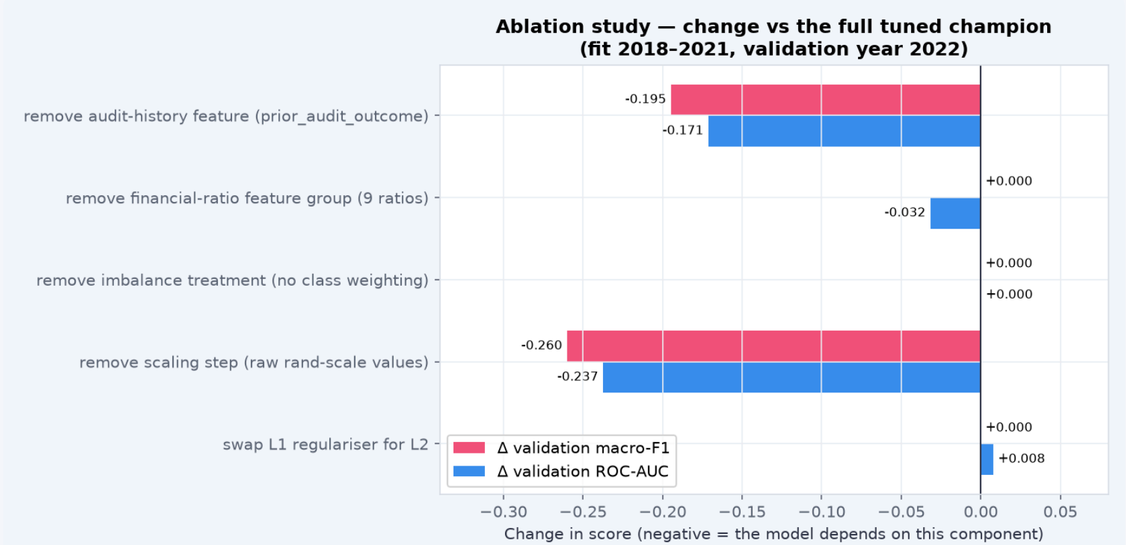
ROC-AUC

rule alone: 0.904

All 22 errors are municipalities whose status changed. The 13 missed were clean in 2023 and failed in 2024; the 9 false alarms improved. The model generalises well but cannot see a change coming.

09 · ABLATION AND MCNEMAR'S TEST

It depends on last year's audit and on scaling



Five one-change experiments, same split and seed. Removing all nine ratios leaves macro-F1 unchanged but lowers ROC-AUC by 0.032.

Champion vs MLP (240 cases)

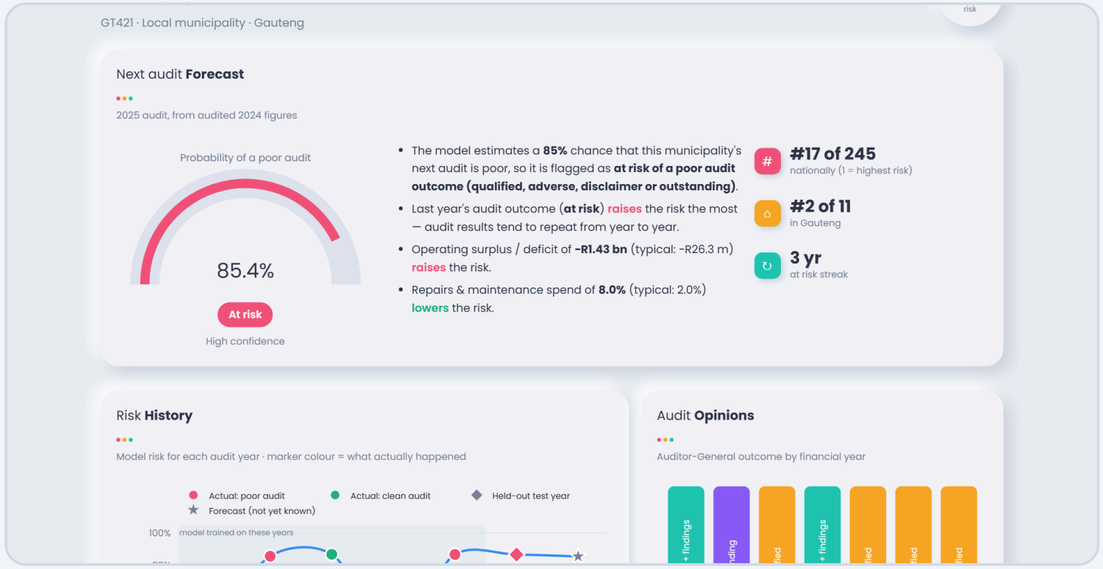
217 both correct	**1** champion only
0 MLP only	**22** both wrong

H_0: equal error rates, $\alpha = 0.05$. One discordant pair, so exact binomial: **p = 1.000, not rejected.**

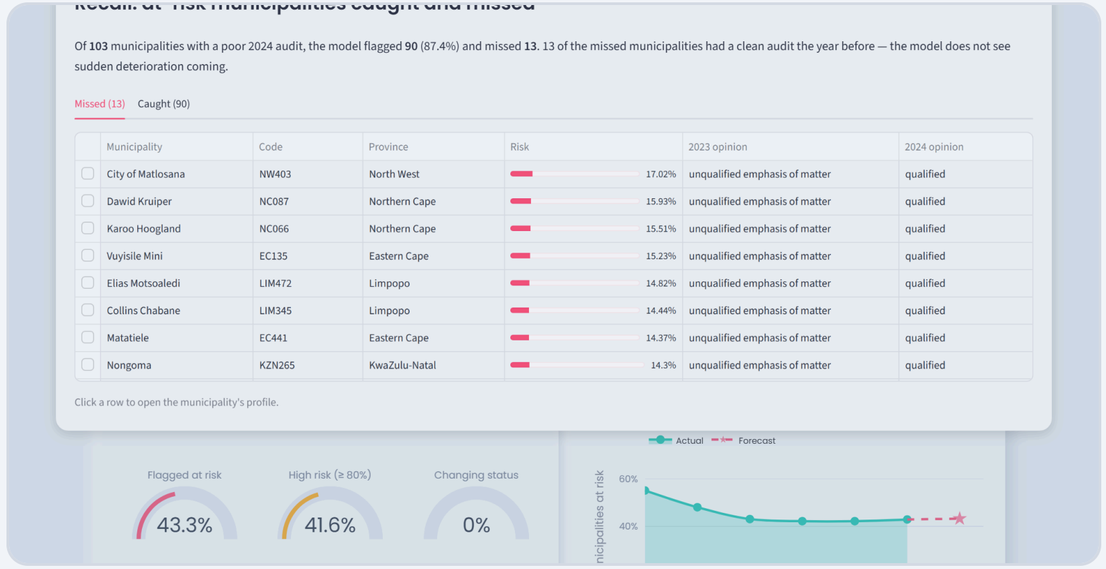
No real difference: we keep Logistic Regression for simplicity and stability.

10 · DEPLOYMENT

A dashboard Treasury can drill into, municipality by municipality



Profile: 2025 forecast with reasons, rank, risk history and audit opinions



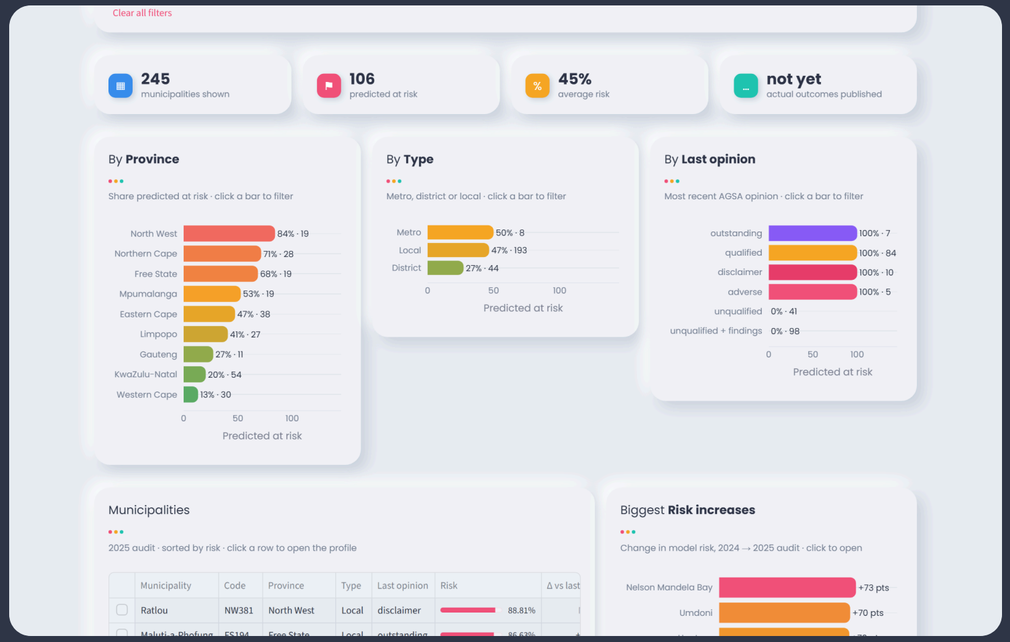
Every KPI opens its detail: here, the 13 municipalities recall missed

Streamlit app running the saved pipeline: overview, explorer, a profile for each of 245 municipalities, risk check with input validation, model evidence and a responsible-use page.

11 · LIVE DEMONSTRATION

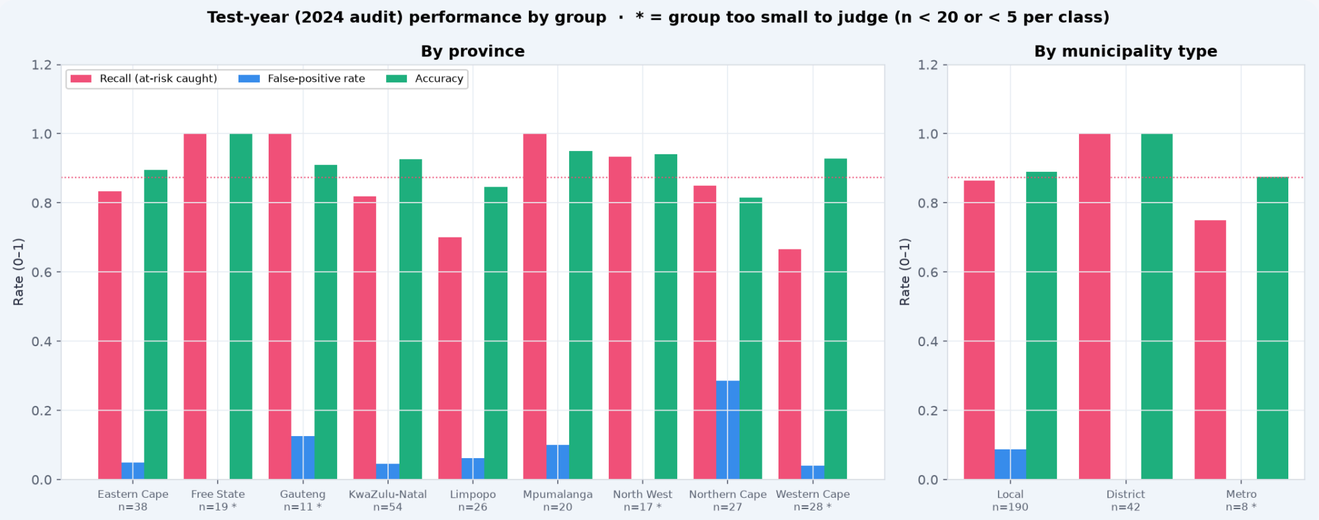
Three valid cases, one invalid file, one drill-down

1. Overview → Recall **Details**: the 13 missed municipalities
2. Top-10 bar → **Ratlou** profile: 89% risk, #1 of 245, 7-year streak
3. Explorer → click **North West**: 16 of 19 flagged
4. Upload **valid_cases.csv**: Masilonyana 89.9%, Tsantsabane 77.2%, eThekwini 9.2%
5. Upload **invalid_case.csv**: three clear errors, nothing scored




```
streamlit run app/app.py
```

Where the model is weakest



Province and type are not model inputs; groups marked * are too small to judge.

0 of 22

status changes predicted correctly; about 22 more are likely in 2025

28.6%

false alarms in the Northern Cape, vs 5–10% elsewhere

70%

recall in Limpopo: 3 of 10 failing municipalities missed

13 · RECOMMENDATION FOR NATIONAL TREASURY

A triage list, not an early warning

Do

- Prioritise the 106 municipalities flagged for 2025 (102 at 80% or more)
- Review each province's list before allocating support
- Use the profiles to choose the kind of support

Do not

- Allocate funding or interventions automatically
- Judge or penalise municipalities or officials
- Rely on it to warn that a clean municipality will fail
- Use it after 2025 without retraining

Monitor yearly

- Compare the forecast with the published 2025 outcomes
- Retrain if macro-F1 falls below 0.866 or recall below 0.80
- Check data quality on every new extract

Next version: add quarterly in-year section 71 data, audit findings and senior vacancies, and predict a **change** in status directly.

14 · REPRODUCIBILITY AND TEAM

Thank you. Questions?

Reproducible end to end

```
python run_all.py
```

11 steps in 4.1 minutes from a fresh clone, with every result identical. Seed 42, pinned requirements, no manual steps or secrets; one notebook per question in notebooks/; dataset card, data dictionary and experiment log in docs/.

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Data governance and preparation (Q2)

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Optimisation, ablation, McNemar (Q5–6)

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Problem definition and evaluation (Q1, Q4)

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Deployment, uncertainty, recommendation (Q7–8)

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Modelling (first Q3 run)

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Data: National Treasury, Municipal Money (municipaldata.treasury.gov.za). AI-assisted code and drafting declared in the report and contribution statement.